Analysis of Author Collaboration Network and Exploration of Link Prediction Models

Vivek Kumar 22372
Dept. of EECS, IISER Bhopal
Bhopal, India
vivek22[at]iiserb[dot]ac[dot]in

Manjul Chaturvedi 22194
Dept. of Data Science, IISER Bhopal
Bhopal, India
manjul22[at]iiserb[dot]ac[dot]in

Shrey Srivastava 22303
Dept. of Data Science, IISER Bhopal
Bhopal, India
shrey22[at]iiserb[dot]ac[dot]in

Harsh Shukla 22354
Dept. of Data Science, IISER Bhopal
Bhopal, India
harshs22[at]iiserb[dot]ac[dot]in

Abstract

In this project, we explore how well different models can predict collaborations in the OGBL-Collab co-authorship graph. We begin by studying the structure of the graph, then experiment with embedding methods, classical machine learning classifiers, and GNN-based link prediction models. Since this is an exploratory class project, we also include unsuccessful attempts, heuristics we tried, and insights gained, especially regarding the difficulties introduced by the temporal nature of collaborations.

1 Introduction

This report presents our exploration of collaboration prediction on the OGBL-Collab dataset. The goal is not to produce a polished research paper but a clear class project report documenting:

- dataset understanding and transformations,
- classical graph analytics,
- shallow embeddings such as DeepWalk and Node2Vec,
- supervised machine learning models,
- GNN-based deep link prediction approaches,
- challenges posed by temporal information.

We structured our project into four main modeling levels and one potential future extension.

GitHub Repository

https://github.com/vickvey/author-collab

Submitted To

Dr. Akash Anil, Dept. of Data Science and Engineering, IISER Bhopal.

2 Dataset Overview

The OGBL-Collab dataset contains approximately 1.1M temporal collaboration edges between about 230k authors. We constructed four versions of the dataset: D0, D1, D2, and D3 for different downstream tasks.

Dataset	Columns	Purpose
D0	a_id, b_id	Unweighted graph
D1	a_id, b_id, weight	Weighted static graph
D2	a_id, b_id, label	Supervised link prediction
D3	a_id, b_id, weight, year	Temporal graph

Table 1: Datasets used in our project.

3 Methodology Overview

To structure the project, we divided our experiments into four levels.

3.1 Level 0: Classical Graph Analytics

Before moving to embedding-based and learning-based models, we first examine the raw structural properties of the co-authorship network. All analyses in this subsection are performed on the D0 view of the dataset, which contains only the unweighted edges `(a_id, b_id)`.

Basic Graph Statistics (D0). The D0 graph is large and extremely sparse, consisting of:

Metric	Value	Comment
Number of nodes	235868	All authors in the training graph
Number of edges	967632	Unique co-authorship pairs
Average degree	8.20	Skewed toward lower-degree authors
Minimum degree	1	No isolated nodes
Maximum degree	382	Presence of prolific hub authors
Density	3.47×10^{-5}	Extremely sparse
Sparsity	0.999965	Expected for academic networks
Connected components	621	Mostly small fragments
Largest component size	232865	Nearly 99 percent of all nodes

Table 2: Basic statistics for the D0 co-authorship network.

Overall, the graph exhibits a classic giant component structure: almost the entire author population is indirectly connected despite very low density.

Degree Distribution and Assortativity. The degree distribution is right skewed. Most authors have few collaborators, while a long tail contains highly prolific researchers.

Statistic	Value	Interpretation
Mean degree	8.20	Typical for collaboration networks
Median degree	4	Most authors have low degree
Standard deviation	13.30	Heavy-tailed distribution
25th, 50th, 75th percentiles	3, 4, 8	Clear right skew
Max degree	382	Highly prolific authors exist
Degree assortativity	0.27	High-degree authors tend to connect together

Table 3: Degree and assortativity statistics for D0.

A positive assortativity coefficient indicates that well-connected authors tend to collaborate with each other. This aligns with known patterns in scientific networks.

Clustering, Motifs, and Community Structure. The D0 network shows very strong clustering behavior. Local clustering is high, indicating strong triadic closure.

Metric	Value	Comment
Average local clustering	0.729	Strong triadic closure
Global clustering	0.363	Dense local groups
Triangles (estimated)	~1.05M	High triangle count
Two paths	55.68M	Large base for triadic closure
Number of communities	831	Many field-specific clusters
Largest community	16487	Likely a large discipline
Modularity	0.8657	Very strong community structure

Table 4: Clustering and community statistics for D0.

The modularity value is exceptionally high, indicating that research fields form cohesive clusters with limited cross-field interaction.

Spectral Analysis. Using the Laplacian spectrum of the largest connected component:

- Algebraic connectivity (λ_2): **0.4216**

A small but non-zero value suggests that the graph is globally connected but contains bottlenecks between communities.

Remarks on D1, D2, D3. **D1 (Weighted View)** Weighted analytics will be performed later. Weighted degrees, clustering, and modularity are expected to shift compared to D0.

D2 (Labeled Edges) D2 is curated for supervised link prediction. Structural analytics do not apply directly.

D3 (Temporal View) D3 preserves year-wise collaboration events. Temporal analysis is introduced later.

4 Level 1: Shallow Encoders and Similarity

In this level, we used shallow graph embedding models to learn low-dimensional node representations and performed link prediction using cosine similarity.

Datasets.

- D0 for DeepWalk.
- D1 for Node2Vec.
- D2 for evaluation using labeled edges.

Embedding Generation.

- Both models used 64-dimensional embeddings.
- DeepWalk used uniform random walks.
- Node2Vec used biased random walks to capture structural and homophily patterns.

Link Prediction Procedure. For each pair (u, v) from D2:

$$\text{sim}(u, v) = \frac{z_u \cdot z_v}{\|z_u\| \|z_v\|}$$

The similarity score is passed through a sigmoid function and thresholded.

Purpose.

- Provide unsupervised baselines.
- Measure predictive strength of node proximity.
- Motivate supervised models due to limitations of similarity scoring.

Results.

- DeepWalk similarity accuracy: **0.3149**.
- Node2Vec similarity accuracy: **0.6946**.

Node2Vec captures more informative structure than DeepWalk.

5 Level 2: Node2Vec Embeddings with Classical Machine Learning Models

We extend link prediction beyond similarity by using Node2Vec embeddings with supervised learning. Node2Vec was trained to produce 64 dimensional embeddings.

Edge features were produced using the Hadamard operator:

$$\phi(u, v) = z_u \odot z_v$$

Models Considered.

- Logistic Regression
- Multilayer Perceptron
- Random Forest
- XGBoost

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.9131	0.9403	0.7748	0.8496
MLP	0.9027	0.9211	0.7614	0.8334
Random Forest	0.9284	0.9538	0.8082	0.8750
XGBoost	0.9225	0.9482	0.7950	0.8635

Table 5: Performance of classical machine learning models trained on Hadamard-transformed Node2Vec embeddings.

Results.

Interpretation.

- Random Forest performed best overall due to its robustness and ability to model nonlinear patterns.
- XGBoost performed similarly well and is effective on tabular features.
- Logistic Regression showed strong linear separability in the Node2Vec space.

- MLP performed reasonably but was less stable without heavy tuning.

6 Level 3: Deep GNN Models

In the final stage of our project, we implemented GCN-based link prediction models following the OGBL-Collab evaluation protocol. All experiments use the standard OGB negative-sampling procedure. Specifically, we rank each true collaboration among a set of 100,000 randomly sampled negative collaborations and compute the fraction of positives that appear within the top-K ranked scores (Hits@K). We use $K = 50$, which is also the setting recommended by the OGBL authors for this dataset. This differs from simpler "top-50 accuracy" style metrics commonly used in other literature because OGBL evaluates ranking quality against a very large negative set, making Hits@K a much stricter and more informative measure of link prediction performance. We follow the official OGB implementation for this.

We evaluate three input-feature configurations for the GCN:

(1) **Node2Vec + GCN**: Pretrained 64-d Node2Vec embeddings used as node features.
(2) **Word2Vec + GCN**: Word2Vec embeddings trained on random-walk sequences of author IDs.
(3) **Concatenated Node2Vec + Word2Vec + GCN**: A combined embedding giving the GCN access to complementary structural signals.

The following results follow the official OGBL-Collab split and evaluator.

Results

Node2Vec + GCN.

- Best Validation Hits@50: **0.5342**
- Test Hits@50: **0.5114**
- Train Hits@50 (sampled negatives): **0.4554**

Word2Vec + GCN.

- Best Validation Hits@50: **0.4563**
- Test Hits@50: **0.4472**
- Train Hits@50 (sampled negatives): **0.3398**

Concatenated (Node2Vec + Word2Vec) + GCN.

- Best Validation Hits@50: **0.5417**
- Test Hits@50: **0.5182**
- Train Hits@50 (sampled negatives): **0.3761**

Discussion

GCN models consistently outperform shallow baselines, validating the usefulness of message passing for collaboration networks. Node2Vec features provide the strongest standalone performance, while concatenating Word2Vec and Node2Vec embeddings yields the best overall results, indicating that the GCN benefits from richer and more diverse input representations.

7 Temporal Challenge (Level 4, Tentative)

A major challenge is the temporal nature of collaborations:

- Older but frequent collaborations dominate similarity scores.
- Recent collaborations should be more relevant.

Possible directions:

- Temporal decay factors
- Dynamic GNNs such as TGAT or TGN
- Year-aware negative sampling

8 Conclusion

This project explored multiple modeling pipelines on OGBL-Collab, focusing on experimentation rather than achieving state-of-the-art performance. Future work includes temporal models and incorporation of metadata.

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